



EXAM PAPERS PRACTICE

Boost your performance and confidence with these topic-based exam questions

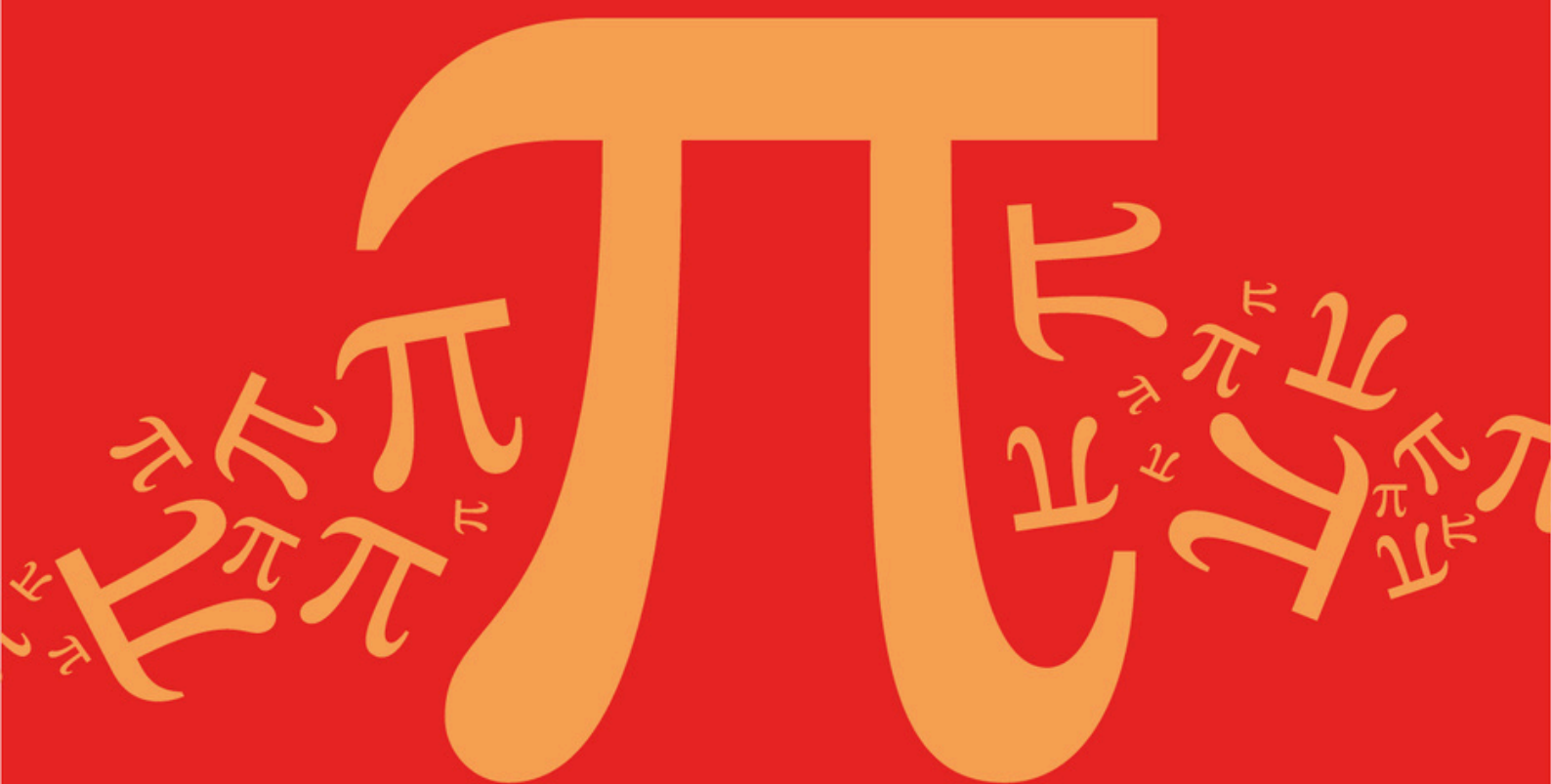
Practice questions created by actual examiners and assessment experts

Detailed mark scheme

Suitable for all boards

Designed to test your ability and thoroughly prepare you

AQA A Level Further Mathematics (7367)



Topic

C: Matrices

Sub topic

C7: Inverse Matrices to Solve Simultaneous Equations

Mark Scheme

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Mark schemes

Q1.

(a) $A^2 = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$

$$= \begin{bmatrix} -1 & -2 & -4 \\ 5 & 6 & 4 \\ 10 & 10 & 9 \end{bmatrix}$$

$p = -1$

p-value

B1

$q = 10$

q-value

B1

2

(b) $A^3 - 6A^2 + 11A - 6I = 0$ multiply by A^{-1}

$(A^3 - 6A^2 + 11A - 6I)A^{-1} = (0)A^{-1}$

Multiplication by A^{-1}

M1

$A^3A^{-1} - 6A^2A^{-1} + 11AA^{-1} - 6IA^{-1} = 0$

$A^2 - 6A + 11I - 6A^{-1} = 0$

$6A^{-1} = A^2 - 6A + 11I$

$A^{-1} = \frac{1}{6}(A^2 - 6A + 11I)$

AG

A1

2

(c) $A^{-1} = \frac{1}{6} \begin{bmatrix} 4 & -2 & 2 \\ -1 & 5 & -2 \\ -2 & -2 & 2 \end{bmatrix}$

$r = 4$

r-value

B1

$s = -2$

s-value

B1

2

(d)
$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 4 & -2 & 2 \\ -1 & 5 & -2 \\ -2 & -2 & 2 \end{bmatrix} \begin{bmatrix} k \\ 5 \\ 7 \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 4k - 10 + 14 \\ -k + 25 - 14 \\ -2k - 10 + 14 \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 4k + 4 \\ 11 - k \\ 4 - 2k \end{bmatrix}$$

use of $A^{-1}v$ – one row correct

M1

$$x = \frac{2k+2}{3}, y = \frac{11-k}{6}, z = \frac{2-k}{3}$$

correct solution for one variable

all correct CAO

A1

A1

Alternative:

If solving equations by elimination, M1 A1 for correct solution for one variable, A1 all correct

3

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Q2.

(a)
$$\left[\begin{array}{ccc|c} 2 & -1 & -1 & 3 \\ 1 & 2 & -3 & 4 \\ 2 & 1 & a & b \end{array} \right]$$

$$\begin{array}{l} r_3 \rightarrow r_3 - r_1 \\ r_2 \rightarrow 2r_2 - r_1 \end{array} \quad \left[\begin{array}{ccc|c} 2 & -1 & -1 & 3 \\ 0 & 5 & -5 & 5 \\ 0 & 2 & a+1 & b-3 \end{array} \right]$$

Correct row operations used to create two zeros in first column – coefficients must be correct

M1

$$r_3 \rightarrow 5r_3 - 2r_2 \quad \left[\begin{array}{ccc|c} 2 & -1 & -1 & 3 \\ 0 & 5 & -5 & 5 \\ 0 & 0 & 5a+15 & 5b-25 \end{array} \right]$$

Use of row operations to create third zero in second column or compare ratios of coefficients in rows 2 and 3

M1

No unique solution: $5a + 15 = 0$

$$a = -3$$

Solves equation with coefficient of $z = 0$ or equation formed from comparison of ratios (eg $a + 1 = -2$) $a = -3$

is a printed answer

A1

3

Alternative 1:

$$2x - y - z = 3 \quad (1)$$

$$x + 2y - 3z = 4 \quad (2)$$

$$2x + y + az = b \quad (3)$$

$$(1) + (3) \Rightarrow 4x + (a - 1)z = b + 3 \quad (4)$$

$$2(3) - (2) \Rightarrow 3x + (2a + 3)z = 2b - 4 \quad (5)$$

Correct elimination of 1 variable.

Coefficients must be correct.

(M1)

$$4(5) - 3(4) \Rightarrow (5a + 15)z = 5b - 25$$

Correctly reduce to one equation with a, b

(M1)

$$5a + 15 = 0$$

$$a = -3$$

Solves equation with coefficient of $z = 0$

(A1)

(3)

Alternative 2:

$$\text{Solve } \begin{vmatrix} 2 & -1 & -1 \\ 1 & 2 & -3 \\ 2 & 1 & a \end{vmatrix} = 0$$

$$2 \begin{vmatrix} 2 & -3 \\ 1 & a \end{vmatrix} - 1 \begin{vmatrix} -1 & -1 \\ 1 & a \end{vmatrix} + 2 \begin{vmatrix} -1 & -1 \\ 2 & -3 \end{vmatrix} = 0$$

Correct expansion by row or column

(M1)

$$2(2a + 3) - (-a + 1) + 2(5) = 0$$

Correct expansion of 2 by 2 determinants

(M1)

$$5a + 15 = 0$$

$$a = -3$$

Solves equation with determinant = 0

(A1)

(3)

- (b) Either $5b - 25 \neq 0$ or $5b - 25 = 0$
Sets their constant $\neq 0$ (or 0)

M1

Inconsistent $b \neq 5$
 CSO (accept $b > 5, b$

A1

2

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Q3.

(a)

$$\begin{vmatrix} 2 & n & 1 \\ 3 & -1 & n \\ -1 & 7 & 1 \end{vmatrix}$$

Expanding the det. of the coeff. mtx.

M1

Setting it = 0

M1

Obtaining & solving a quadratic eqn. in n

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M1

$$0 = n^2 + 17n - 18 = (n + 18)(n - 1) \Rightarrow n = 1, -18$$

CSO

A1

4

- (b) $n = 1$ gives $2x + y + z = 5$
 $3x - y + z = 1$
 $-x + 7y + z = 1$
ft their chosen integer n

B1

Eliminating one variable from a pair of equations, twice

M1

e.g. $(2) - (1) \Rightarrow x - 2y = -y = -4$

A1 ft

and $(2) - (3) \Rightarrow 4x - 8y = 0$

A1 ft

Inconsistency clearly demonstrated from fully correct working

E1

3 planes have no common intersection (or form a Δ^r prism)

*Also ft "3 planes meet in a common line" or
"3 planes form a sheaf" if consistency conclusion made*

B1 ft

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Examiner reports

Q1.

This question elicited a good response from students. There was evidence of efficient use of graphic calculators in parts (a) and (c). In part (b), students had to produce a watertight proof to get both marks. It was disconcerting to find students writing “divide by A” or “cancel A”, displaying a lack of understanding matrix algebra. In the final part, many chose not to use the inverse matrix to find the solutions and instead chose to use elimination. Varying degrees of success were seen – the most successful chose to eliminate y from the second and third equations.

Q2.

The most successful method in part (a) was to find the determinant and set it to equal zero. The least successful approach was to try to combine equations in a more traditional manner. Students simply got themselves lost in the algebra and could not complete the method. The row operation method of elimination was rarely seen but often proved successful and helped with part (b). Part (b) illustrated again that students are not confident or clear about consistent and inconsistent equations readily getting them mixed up.

Q3.

This was another generally popular and successful question for candidates. Most attempts scored all 4 marks on part (a) and most of those for part (b). However, the approaches taken often provided further evidence that the candidature, in general, was not at the appropriate level of mathematical maturity required for taking their MFP4 paper at this time. Having supplied three equations with a common coefficient of 1 for z , easing the work needed to establish the inconsistency of the system, most candidates preferred to work with the x 's and y 's instead, which just took a little longer and led more often to the introduction of numerical errors. Also, it was often the case that inconsistency was “established” on the basis that two constants weren't equal, when comparing a couple of equations, but candidates' lack of care in the arithmetic frequently meant that one or other of these two constants wasn't the correct one, which invalidated their conclusion and lost them a mark.